

Question Number	Answer	Additional Guidance	Mark
2a	• suitable risk clearly identified (1) • reduction of risk identified to the scientist (1)	Candidates will express this in a number of different ways e.g. breaking fibre damages eye so use goggles / safety screen; allergies to plant / fibre so wear gloves; hydrogen peroxide is irritant so wear gloves; falling masses may cause injury so place a cushion beneath / stand back	2

Question Number	Answer	Additional Guidance	Mark
2b	• correct mean calculated (1) • correct percentage increase (1) • correct answer to 3 sig figs (1)	ECF for using incorrect means for MP2 and 3 834 10.432 10.4 Correct answer only gains 3 marks	3

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2ci	An answer that includes two of the following points: Abiotic • pH (of soil) (1) • water content (of soil) (1) • organic matter (of soil) (1) • concentration of hydrogen peroxide (1) Biotic • age of { plant / fibre} (1) • species (1) • {diameter / length} of fibre (1)	Allow humidity (in room) Allow mineral content (of soil), ignore nutrients Allow variety / cultivar / type Allow thickness	2

Question Number	Answer	Additional Guidance	Mark
2cii	• description of an appropriate method of control (1)	Allow control of temperature or other unsuitable variable If water / pH selected it must be controlled in soil	1

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2d	A description that includes two of the following: • (long) cells with {secondary thickening / lignin} (1) • end wall present (1) • {dead cells / no cell contents / hollow} (1) • pits present (1)		2

Total for question 2 = 10 marks)

